# STA2001 Probability and Statistics I

Assignment 10

Deadline: 23:59 pm, Dec 5th, Friday

1. The pdf of $X$ is $f(x) = 2x$, $0 < x < 1$.
   (a) Find the cdf of $X$.
   (b) Describe how an observation of $X$ can be simulated.
   (c) Simulate 10 observations of $X$ (provide the codes).

2. Let $X_1, X_2, X_3$ be three independent random variables with binomial distributions $b(4, 1/2)$, $b(6, 1/3)$, and $b(12, 1/6)$, respectively. Find
   (a) $P(X_1 = 2, X_2 = 2, X_3 = 5)$;
   (b) $E[X_1 X_2 X_3]$;
   (c) The mean and the variance of $Y = X_1 + X_2 + X_3$.

3. Let $X_1, X_2, X_3$ denote a random sample of size $n = 3$ from a distribution with the geometric pmf

$$f(x) = (0.25)^{x-1}(0.75) \quad x = 1, 2, \ldots.$$

   (a) Compute $P(X_1 = 1, X_2 = 3, X_3 = 1)$.
   (b) Determine $P(X_1 + X_2 + X_3 = 5)$.
   (c) If $Y$ equals the maximum of $X_1, X_2, X_3$, find $P(Y \leq 2)$.

4. Let $X_1, X_2, X_3$ be independent random variables that represent lifetimes (in hours) of three key components of a device. Say their respective distributions are exponential with means 1000, 1500, and 2000. Let Y be the minimum of $X_1, X_2, X_3$ and compute $P(Y > 1000)$.

5. Let $X_1$ and $X_2$ have independent distributions $b(n_1, p)$ and $b(n_2, p)$. Find the mgf of $Y = X_1 + X_2$. How is $Y$ distributed?

6. Let $X_1, X_2, X_3, X_4, X_5$ be a random sample of size 5 from a geometric distribution with $p = 1/3$.

(a) Find the mgf of $Y = X_1 + X_2 + X_3 + X_4 + X_5$.

(b) How is $Y$ distributed?

7. Let $X_1$ and $X_2$ be two independent random variables. Let $X_1$ and $Y = X_1 + X_2$ be $\chi^2(r_1)$ and $\chi^2(r)$, respectively, where $r_1 < r$.

(a) Find the mgf of $X_2$.

(b) What is its distribution?

8. Let $X$ be $N(0, 1)$. Use the mgf technique to show that $Y = X^2$ is $\chi^2(1)$. Hint: Evaluate the integral representing $E[e^{tX^2}]$ by writing $w = x\sqrt{1 - 2t}$.

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T1. (a.) when $x\le 0$: $F(x)=0$; when $x>1$: $F(x)=1$. when $0<x\le 1$: $\int_0^x 2t\,dt = t^2\Big]_0^x = x^2$

$$\Rightarrow F(x)=\begin{cases}0, & x\le 0\\ x^2, & 0<x\le 1\\ 1, & x>1\end{cases}$$

(b.) generate $y$ from $Y\sim U(0,1)$, then let $y=F(x) \Rightarrow y=x^2 \Rightarrow x=\sqrt{y}$, $x$ is a observation of $X$.

(c.)

```python
import numpy as np

# 生成10个均匀分布(0,1)的随机数
u = np.random.uniform(0, 1, size=10)
# 逆变换: x = sqrt(u)
x_obs = np.sqrt(u)

print("10个X的观测值:")
print(x_obs)
```

$\Rightarrow$

```
10个X的观测值:
[0.93881676 0.95030377 0.63336127 0.94588012 0.15433707 0.24243996
 0.36070096 0.22477168 0.60164886 0.75358269]
```

T2. (a.) $P(X_1=2,X_2=2,X_3=5)=P(X_1=2)\cdot P(X_2=2)\cdot P(X_3=5)=C_4^2\cdot(\frac{1}{2})^4\cdot C_6^2\cdot(\frac{1}{3})^2\cdot(\frac{2}{3})^4\cdot C_{12}^5\cdot(\frac{1}{6})^5\cdot(\frac{5}{6})^7=\frac{3}{8}\times\frac{80}{243}\times 792\times\frac{1}{7776}\times(\frac{5}{6})^7\approx\frac{3}{8}\times\frac{80}{243}\times 0.035\approx 0.0045$

(b.) $E(X_1X_2X_3)=E(X_1)\cdot E(X_2)\cdot E(X_3)=4\times\frac{1}{2}\times 6\times\frac{1}{3}\times 12\times\frac{1}{6}=8$

(c.) $Y=X_1+X_2+X_3$; $E(Y)=E(X_1)+E(X_2)+E(X_3)=4\times\frac{1}{2}+6\times\frac{1}{3}+12\times\frac{1}{6}=2+2+2=6$

$Var(Y)=Var(X_1)+Var(X_2)+Var(X_3)=4\times\frac{1}{2}\times(1-\frac{1}{2})+6\times\frac{1}{3}\times\frac{2}{3}+12\times\frac{1}{6}\times\frac{5}{6}=1+\frac{4}{3}+\frac{5}{3}=4$

T3. (a.) $P(X_1=1,X_2=3,X_3=1)=P(X_1=1)P(X_2=3)P(X_3=1)=0.75\times(0.25)^2\times 0.75\times 0.75=0.026367187 5\approx 0.026$

(b) $X\sim Geometric(0.75)$, $S=X_1+X_2+\cdots+X_n\sim$ Negative Binomial$(3,0.75)$

$P(S=5)=\binom{5-1}{3-1}(0.75)^3(1-0.75)^{5-3}=6\times 0.421875\times 0.0625=0.158203125\approx 0.1582$

(c.) $P(Y\le 2)=P(X_1\le 2,X_2\le 2,X_3\le 2)=(P(X_i\le 2))^3$

$\Rightarrow P(X_i\le 2)=P(X_i=1)+P(X_i=2)=0.75+(0.25)^{2-1}\times 0.75=0.75+0.1875=0.9375$

$\therefore P(Y\le 2)=(0.9375)^3=0.824218 75\approx 0.824$

T4. $P(Y>1000)=P(X_1>1000)\cdot P(X_2>1000)\cdot P(X_3>1000)$,

$P(X_i>t)=1-F_i(t)=e^{-\frac{t}{\mu_i}}$, $i=1,2,3$ $\quad \mu_1=1000,\ \mu_2=1500,\ \mu_3=2000$

$\Rightarrow P(Y>1000)=e^{-1}\cdot e^{-\frac{2}{3}}\cdot e^{-\frac{1}{2}}=e^{-\frac{13}{6}}\approx 0.105$

T5. Independent $\Rightarrow M_Y(t)=M_{X_1}(t)\cdot M_{X_2}(t)=(pe^t+1-p)^{n_1}\cdot(pe^t+1-p)^{n_2}=(pe^t+1-p)^{n_1+n_2}\Rightarrow Y\sim b(n_1+n_2,p)$.

T6. $X_i\sim Geom(\frac{1}{3})$, $\Rightarrow M_{X_i}(t)=\frac{\frac{1}{3}e^t}{1-\frac{2}{3}e^t}$, $t<\ln\frac{3}{2}$

$M_Y(t)=M_{X_1}(t)\cdot M_{X_2}(t)\cdots M_{X_5}(t)=(M_{X_i}(t))^5=\frac{(\frac{1}{3})^5e^{5t}}{(1-\frac{2}{3}e^t)^5}\Rightarrow Y\sim$ Negative Binomial$(5,\frac{1}{3})$

T7. (a.) $M_Y(t)=M_{X_1}(t)\cdot M_{X_2}(t)$

$\Rightarrow M_{X_2}(t)=\frac{M_Y(t)}{M_{X_1}(t)}=\frac{\frac{1}{(1-2t)^{\frac{r}{2}}}}{\frac{1}{(1-2t)^{\frac{n}{2}}}}=\frac{1}{(1-2t)^{\frac{r-n}{2}}}$

(b.) $X_2\sim\chi^2(r-n)$

T8. $M_Y(t)=E(e^{tY})=E(e^{tX^2})=\int_{-\infty}^{\infty}e^{tx^2}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}x^2(1-2t)}dx$; Let $w=x\sqrt{1-2t}$; $1-2t>0$, $t<\frac{1}{2}$. $\Rightarrow x=\frac{w}{\sqrt{1-2t}}$, $dx=\frac{1}{\sqrt{1-2t}}dw$.

$\therefore M_Y(t)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{w^2}{2}}\cdot\frac{1}{\sqrt{1-2t}}dw=\frac{1}{\sqrt{2\pi}}\cdot\frac{1}{\sqrt{1-2t}}\cdot\sqrt{2\pi}=\frac{1}{\sqrt{1-2t}}\Rightarrow Y\sim\chi^2(1)$ ∎.